

Nam Nguyen

Senior Data & AI Engineer

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Summary

Results-driven Senior Data & AI Engineer with 11 years of experience delivering scalable data platforms across Microsoft ecosystem consulting and retail analytics domains. Specialized in designing Azure-based lakehouse architectures, building robust data pipelines, and enabling enterprise analytics through optimized dimensional models and API integrations. Proven track record of translating complex business requirements into reliable data solutions using Azure Data Factory, Databricks, and Synapse while improving data quality, automation, and performance for mission-critical systems.

Skills

- **Languages:** SQL, Python, PySpark, T-SQL
- **Libraries/Frameworks:** Apache Spark, Azure Databricks, Pandas, REST API Integration, OAuth 2.0, JSON Processing
- **Data/Databases:** Azure SQL Database, Synapse Analytics, ADLS, Data Warehousing, Star Schema, Dimensional Modeling, Data Lakehouse Architecture, ETL/ELT Pipelines, Data Integration, Metadata-driven Pipelines
- **Cloud/Platforms:** Microsoft Azure, Azure Data Factory, Azure Databricks, Azure Synapse, Azure Fabric, Azure Storage, CI/CD Pipelines, DevOps
- **Soft Skills:** Analytical problem solving, stakeholder communication, solution design, mentoring, documentation, client collaboration, adaptability, ownership mindset

Work Experience

Senior Data Engineer, Vontier

Sep 2023 – Present

- Worked on a connected mobility analytics platform integrating fueling systems, fleet telemetry, and diagnostics data into a unified **Azure Data Lakehouse** using **Azure Data Factory** and **ADLS**, enabling centralized reporting and improving operational visibility across transportation networks.
- Addressed fragmented ingestion pipelines by building scalable **ETL workflows** using **Azure Data Factory** and **PySpark** in **Azure Databricks**, implementing metadata-driven orchestration patterns that reduced pipeline maintenance complexity and improved deployment consistency.
- Resolved inconsistencies in third-party API ingestion by developing robust **REST API** connectors with **OAuth 2.0** authentication and advanced **JSON normalization** techniques, ensuring reliable ingestion into relational structures for downstream analytics.
- Improved reporting performance by designing optimized **star schemas** in **Synapse Analytics**, applying dimensional modeling techniques to reduce query latency and enhance business intelligence usability across multiple reporting tools.
- Enhanced data reliability by implementing validation frameworks within **Azure Databricks** pipelines using **PySpark**, ensuring schema consistency and reducing data quality issues across high-volume ingestion processes.
- Modernized legacy data workflows by transitioning batch pipelines into scalable **lakehouse architecture** leveraging **ADLS** and **Synapse**, improving data accessibility while maintaining cost efficiency and governance standards.
- Built reusable pipeline templates using **Azure Data Factory** and parameterized configurations, allowing faster onboarding of new data sources and improving development velocity across multiple data projects.
- Supported near real-time data processing requirements by integrating streaming ingestion patterns using **Databricks Spark**, enabling faster insights for fleet monitoring use cases.
- Designed modular data transformation layers using **SQL** and **PySpark**, enabling scalable transformations and simplifying downstream reporting dependencies across multiple business domains.
- Contributed to system architecture decisions by implementing layered data zones within **Azure Data Lake**, improving data governance, traceability, and lifecycle management.
- Configured CI/CD pipelines using **Azure DevOps** to automate deployment of **Data Factory** and **Databricks** assets, reducing manual errors and ensuring consistent environment promotion.
- Collaborated with cross-functional teams in Agile environments to translate business requirements into technical implementations, ensuring alignment between analytics goals and data engineering deliverables.
- Mentored junior engineers by guiding best practices in **SQL optimization**, **pipeline design**, and **Azure services**, improving team productivity and solution consistency.

Senior Data Engineer, Callaway Golf

Aug 2022 – Jun 2024

- Developed a product analytics platform aggregating equipment usage and e-commerce interactions into centralized storage using **Azure SQL** and **Data Factory**, supporting business insights into customer behavior and product performance.
- Improved ingestion reliability by designing **ETL pipelines** that processed structured and semi-structured data using **Python** and **SQL**, enabling accurate reporting for marketing and sales analytics.
- Optimized backend services by implementing efficient **SQL transformations** and indexing strategies, reducing query execution time and improving dashboard responsiveness.

- Built API integration workflows to ingest third-party retail data using **REST APIs** and **JSON parsing**, ensuring seamless integration into internal reporting systems.
- Designed modular data processing layers that separated ingestion, transformation, and reporting logic, improving maintainability and scalability of the platform.
- Implemented cloud-based storage solutions using **Azure Storage** for handling growing data volumes, ensuring cost-effective scaling and reliable data retention.
- Automated deployment processes using CI/CD pipelines, improving release consistency and reducing manual configuration issues.
- Collaborated within Agile teams to align engineering efforts with product requirements, ensuring timely delivery of analytics capabilities.

Data Engineer, Signet Jewelers

Oct 2021 – Aug 2022

- Built a retail analytics pipeline consolidating sales and inventory data using **Azure Data Factory** and **SQL**, enabling consistent reporting across multiple brands and improving decision-making capabilities.
- Improved data transformation efficiency by implementing reusable **SQL-based ETL logic**, reducing redundancy and ensuring standardized data outputs.
- Developed API ingestion workflows using **Python** to integrate vendor data feeds, ensuring timely updates to pricing and product availability datasets.
- Designed data storage strategies using **Azure SQL Database**, improving data accessibility and performance for reporting tools.
- Built a secondary customer insights platform using transactional data and behavioral tracking, supporting targeted marketing initiatives.
- Worked closely with analysts and engineers to deliver reliable datasets aligned with reporting and operational needs.

Data Engineer, Neiman Marcus Group

Jul 2019 - Oct 2021

- Developed data pipelines supporting e-commerce analytics using **SQL** and **Azure Data Factory**, enabling insights into customer purchasing patterns and product performance.
- Improved reporting accuracy by implementing structured **data modeling** techniques and maintaining clean data pipelines.
- Integrated external vendor systems through **API-based ingestion**, ensuring data consistency across multiple business units.
- Optimized backend data workflows by restructuring inefficient queries and improving transformation logic.
- Built a recommendation analytics dataset supporting personalization features in online retail systems.
- Collaborated with teams to deliver data solutions aligned with business needs and system scalability requirements.

Associate Engineer, JCPenney

Jun 2016 - Jul 2019

- Supported development of internal reporting systems using **SQL** and ETL processes, enabling operational visibility across retail operations.
- Maintained backend data systems ensuring reliable data availability for reporting tools and dashboards.
- Developed scripts for automating repetitive data processing tasks, improving operational efficiency.
- Built supplementary reporting datasets for merchandising analysis and inventory tracking.

Data Engineer, Federation of State Medical Boards

May 2015 - Jan 2016

- Developed healthcare data pipelines supporting physician licensing systems using **SQL**, ensuring accurate and timely processing of regulatory data.
- Improved backend processing reliability by implementing structured ETL workflows.
- Built reporting datasets for compliance monitoring and operational insights.
- Collaborated with stakeholders to align data systems with regulatory requirements.

Education

Bachelor's of Computer Science, The University of Texas at Dallas

2011 - 2016